LESSON 6 CURRICULUM DESIGN AND CURRICULUM DESIGN MODELS

6.1 Introduction

Having discussed the dimensions of curriculum in our last lesson, I wish to welcome us to our sixth lesson which is curriculum design. In this lesson we shall discuss the curriculum design and curriculum design models. We all are aware of the term design. Design is a very broad concept and its meaning can greatly vary from one field to another. From the many definitions of the term design, we all agree that a design is a plan or drawing produced to show the look and function or workings of a building, garment, or other object before it is made. A design is a plan or specification for the construction of an object or system or for the implementation of an activity or process or the result of that plan or specification in the form of a prototype, product, or process. Having defined design as such, we could apply the same definition to curriculum and say that curriculum design refers to the structure or organization of the curriculum. Curriculum design refers to the structure or organization of the curriculum, and curriculum development which includes the planning, implementation, and evaluation processes of the curriculum. Just as designs are guided by models, in the same way curriculum design process is guided by curriculum design models. Welcome to our lesson.

6.2 Lesson learning outcomes

By the end of this lesson, you should be able to:

6.2.1 Describe curriculum design and its purpose

6.2.2 Explain the forms of curriculum design

6.2.3 Describe the various curriculum design models

6.2.4 Highlight some of the factors affecting curriculum design

6.2.1 Curriculum design

Curriculum design is the planning period when instructors organize the instructional units for their course. Curriculum design is the organization is the process to change the content into students learning experiences intentionally, and make learning experiences sequential, integral, successive after curriculum ideology has been determined, curriculum goal set, curriculum goal been selected.

Curriculum design is largely concerned with issues such as what to include in the curriculum and how to present it in such a way that the curriculum can be implemented with understanding and success. Curriculum design involves planning activities, readings, lessons, and assessments that achieve educational goals. Curriculum design is concerned with issues of choosing what the organizational basis or structural framework of the curriculum is. The choice of a design often implies a value position. Therefore, curriculum design refers to how the components of the curriculum have been arranged in order to facilitate learning

Curriculum design is a way of organizing that permits curriculum ideas to function. She also adds that curriculum design refers to the structure or pattern of the organization of the curriculum.

The curriculum design process results in a curriculum document that contains the following:

- a statement of purpose(s),
- an instructional guide that displays behavioral objectives and content organization in harmony with school organization,
- a set of guidelines (or rules) governing the use of the curriculum, and
- an evaluation plan.

Thus, curriculum is designed to fit the organizational pattern of the school/institution for which it is intended. How a curriculum is conceptualized, organized, developed, and implemented depends on a particular state's or district's educational objectives. Whatever design is adopted depends also on the philosophy of education.

Teachers design each curriculum with a specific educational purpose in mind. The ultimate goal is to improve student learning, but there are other reasons to employ curriculum design as well. For example, designing a curriculum for high school students with both primary and college/university curricula in mind helps to make sure that learning goals are aligned and complement each other from one stage to the next. If a high school curriculum is designed without taking prior knowledge from primary school or future learning in university or college into account it can create real problems for the students.

6.2.2 Forms of curriculum design

Curriculum design can be segmented or organized into various forms. These include subject-centered design, learner-centered design and problem-centered design, objective and process design models.

6.2.2.1 Subject-centered curriculum design

A model where the curriculum is divided into subject areas, and there is little flexibility for cross-curricular activity. Emphasis is placed on acquisition, memorization, and knowledge of each specific content area. Within this curriculum structure, strong emphasis is placed on instruction, teacher-to-student explanation, and direct strategies. Direct strategies include lectures, questions, and answers, as well as teacher-student discussions. These curricula often encourage memorization and repetitive practice of facts and ideas. Traditionally, students had little choice about what they studied under these curricula. Subject centered designs are by far the most popular and widely used curriculum designs. This is because knowledge and content are well accepted as integral parts of the curriculum.

The following subject centered designs category comprises several examples, i.e.

i. Subject design
ii. Discipline designs
iii. Broad fields designs
iv. Correlations designs

i) Subject design

It is the oldest school design and the best known to both teachers and lay people, because this is what has been employed over years in educating them. It is popular because it corresponds to textbook treatment and how teachers are trained as subject specialists. This design is based on a belief that what makes humans unique and distinctive is there intellect; the searching for and the attainment of knowledge are the natural fulfillment to that intellect. In the subject matter design, the curriculum is organized according to how knowledge has been developed in the various subject areas e.g. Biology, Physics, Geography, English, and Mathematics.

ii) Broad fields designs

The broad fields design, another variation of the subject centered design, appeared as an effort to correct the fragmentation and compartmentalization caused by the subject design. This design permits the melding of the two or more related subjects into a single broad or fused field of study. It is what we term today as integrated studies e.g. Integration of English literature.

One reason for its continued popularity is that it dissolves subject boundaries in ways that makes the information more useful and meaningful to the pupils. It allows the teacher to have more flexibility in choosing content. By integrating separate subjects, it enables, learners to see relationships among the various subjects of the curriculum.

iii) Correlation design

It is a design employed by those who do not wish to go as far as creating a broad fields design, but who do realize that there are times when separate subjects require some linkage in order to reduce fragmentation of curricular content.

Correlation is an attempt to eliminate the isolation and compartmentalization of subjects without radically overhauling the subject curriculum. For example, a Science teacher may collaborate with the social science teacher by having students write papers dealing with the history of some particular scientific theory.

6.2.2.2 Learner-centered curriculum design

In contrast to subject centred curriculum design learner-centered curriculum design takes each individual's needs, interests, and goals into consideration. In other words, it acknowledges that students are not uniform and adjust to those student needs. Learner-centered curriculum design is meant to empower learners and allow them to shape their education through choices.

Instructional plans in a learner-centered curriculum are differentiated, giving students the opportunity to choose assignments, learning experiences or activities. This can motivate students and help them stay engaged in the material that they are learning.

Differentiated instructional plans provide an opportunity to select assignments, teaching and learning experiences, or activities that are timely and relevant. This form of curriculum design has been shown to engage and motivate students. The drawback to this form of curriculum design is

that it can create pressure to form content around the learning needs and preferences of students. These insights can be challenging to glean in a largely online learning environment. Balancing individual student interests with the course's required outcomes could prove to be a daunting task that takes a learner-centered approach to building your curriculum.

Weimer (2002) described five learner centered practice areas that need to change to achieve learner – centered teaching;

- The function of the content
- The role of the instructor
- The responsibility for learning
- The process and purpose or assessment
- The balance of power.

The function of the content in learner-centered teaching include building a strong knowledge foundation and to develop learning skills and learner self-awareness.

The role of the instructor should focus on student learning. The roles are facilitative rather than didactic.

The responsibility of learning shifts from the instructor to the students. The instructor creates learning environments that motivate students to accept responsibility for learning.

The processes and purposes of assessment shift from only assigning grades to include constructive feedback and to assist with improvement.

Learner-centered uses assessment as part of the learning process.

6.2.2.3 Problem-centered curriculum design

Problem centered design focuses on the problems of living on the perceived realities of institutional and group life both for the individual and for society in general. This is a curriculum design that also uses a student approach but that instructs students to look at a problem or situation and figure out a way to solve it.

Teachers expect students to use their real-life experiences to determine an answer teaches students how to look at a problem and formulate a solution. A problem-centered curriculum model helps

students engage in authentic learning because they're exposed to real-life issues and skills, which are transferable to the real world.

Problem-centered curriculum design has been shown to increase the relevance of the curriculum and encourages creativity, innovation and collaboration in the classroom. The drawback to this model is that the individual needs and interests of students aren't always accounted for. By considering all three models of curriculum design before they begin planning, instructors can choose the model that is best suited to both their students and their course.

Problem centered curriculum designs are organized to reinforce cultural traditions and also to address those community and societal needs that are currently unmet. Though these designs place the individual in a social setting, they are unlike learner centered designs in a major way. Problem centered designs are planned before the arrival of students.

6.2.3 Curriculum Design Models

Curriculum components can be organized in numerous ways. This is done using various using curriculum design models. Curriculum model is a broad term referring to the guide used to write curriculum guides, or the documents used in education to determine specific aspects of teaching, such as subject, time frame, and manner of instruction. Curriculum design models are based on a body of theory about teaching and learning. A Curriculum model determines the type of curriculum used it encompasses educational philosophy, approach to teaching, and methodology. Curriculum literature still has many more models for design. There are many other curriculum design models developed by different scholars. Most of these models are objectives-based, i.e. they focus on objectives as the basis upon which the entire design process is based, and draw a lot from the work of Ralph Tyler. Those include the Wheeler, Kerr, and Taba models.

6.2.3.1 The Objectives Model

The objectives model of curriculum design contains content that is based on specific objectives. These objectives should specify expected learning outcomes in terms of specific measurable behaviours. This model comprises four main steps:

- agreeing on broad aims which are analysed into objectives,
- constructing a curriculum to achieve these objectives,

- refining the curriculum in practice by testing its capacity to achieve its objectives, and
- communicating the curriculum to the teachers through the conceptual framework of the objectives.

6.2.3.2 The Process Model

Unlike the objectives model, this model does not consider objectives to be important. Using this model presupposes that:

- Content has its own value. Therefore, it should not be selected on the basis of the achievement of objectives.
- Content involves procedures, concepts and criteria that can be used to appraise the curriculum.
- Translating content into objectives may result in knowledge being distorted.
- Learning activities have their own value and can be measured in terms of their own standard. For this reason, learning activities can stand on their own.

Ralph Tyler's Model

Ralph W. Tyler was an American educator who worked in the field of assessment and evaluation. He served on or advised a number of bodies that set guidelines for the expenditure of federal funds and influenced the underlying policy of the Elementary and Secondary Education Act of 1965. Tyler's curriculum design Model (1949) is based on the following four (4) fundamental questions he posed for guiding the curriculum design process. They are as follows:

1. What educational purposes is the school seeking to attain?
2. What educational experiences are potentially provided that are likely to attain these purposes?
3. How can these educational experiences be effectively organized?
4. How can we determine whether these purposes are being attained?

According to Tylor, in developing any curriculum should involve answering these four questions. Moreover, this model is linear in nature, and objectives form the basic component in the curriculum development. Moreover, objectives are the basis for the selection and organisation of learning experiences, and they form the basis for assessing the curriculum. Furthermore, objectives are

derived from the learner, contemporary life and subject specialist. Thus, it is an **objective model**. Schematically, Tyler's model is presented as follows.

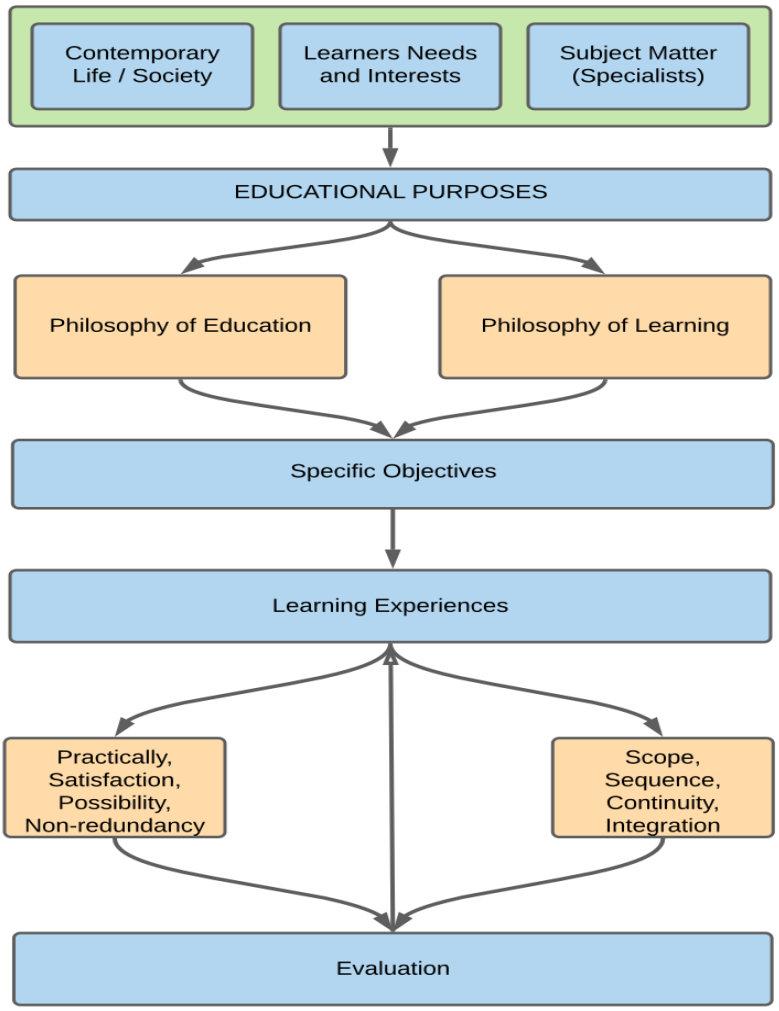


Figure 1: Ralph Tyler's Model

Application of Ralph Tyler's Model in Curriculum Design

In applying Tyler's model to curriculum design, the process begins with framing objectives for the curriculum. Because of its emphasis on the importance of objectives, it is considered an objective-based model. This process starts with analyzing information from various data sources. Data sources for curriculum according to Tyler include:

Contemporary society/life: For this source, the designer analyzes the issues affecting society that could be solved through education. Examples are cultural issues, socio-economic issues, and health issues such as HIV/AIDS among.

- **Learner's needs and interests**
- **Subject specialists/subject matter**

From these sources, the designer develops general objectives. These are subjected to a screening process, using the philosophy of education and psychology of learning as the major screens. Social values are also used as a screen, but sometimes these are subsumed in the philosophy of education. This yields a feasible number of objectives that are focused on in education.

Specific objectives are then derived from the general objectives. For each of the specific objectives, learning experiences are identified. In this context, the learning experiences include the subject matter/content and learning activities.

The next step is the organization of learning experiences. This is done to ensure effective learning takes place. The various principles of the organization include scope, sequence, integration, and continuity, among others. The final step involves evaluation, to determine the extent to which the objectives have been met.

Feedback from the evaluation is then used to modify the learning experiences and the entire curriculum as found necessary.

Learning Experiences

Learning experiences refer to the interaction between the learner and the external conditions in the environment which they encounter. Learning takes place through the active participation of the students; it is what the students are involved in that they learn, not what the teacher does.

The problem of selecting learning experiences is the problem of determining the kind of experiences likely to produce given educational objectives and also the problem of how to set up opportunity situations that evoke or provide within the student the kinds of learning experiences desired.

General Principles in Selecting Learning Experiences

- Provide experiences that give students opportunities to practice the behavior and deal with the content implied.
- Provide experiences that give satisfaction from carrying on the kind of behavior implied in the objectives.
- Provide experiences that are appropriate to the student's present attainments, his/her predispositions.
- Keep in mind that many experiences can be used to attain the same educational objectives.
- Remember that the same learning experience will usually bring about several outcomes.

Criticisms leveled at the Tyler's Model

There are several criticisms leveled at the Tyler's goal attainment model or the Tyler's objective centered model. The first criticism is that, it is difficult and time consuming to construct behavioral objectives. Tyler's model relies mainly on behavioral objectives.

The objectives in Tyler's model comes from three sources (the student, the society, and the subject matter) and all the three sources have to agree on what objectives needs to be addressed. This is a cumbersome process. Thus, it is difficult to arrive to consensus easily among the various stakeholders' groups.

The second criticism is that, it is too restrictive and covers a small range of student skills and knowledge.

The third criticism is that Tyler's model is too dependent on behavioral objectives and it is difficult to declare plainly in behavioral objectives the objectives that covers none specific skills such as those for critical thinking, problem solving, and the objectives related to value acquiring processes.

The fourth and last criticism is that the objectives in the Tyler's model are too student centered and therefore the teachers are not given any opportunity to manipulate the learning experiences as they see fit to evoke the kind of learning outcome desired.

John Goodlad's Model

John Goodlad was an educational researcher and theorist who published influential models for renewing schools and teacher education. The Goodlad model deviates a bit from the Ralph Tyler's model. It is particularly unique in its use of social values. Whereas Tyler considers them as a screen, Goodlad proposes they are used as data sources. Hence, Goodlad proposes four data sources:

- values,
- funded knowledge,
- conventional wisdom, and
- students' needs and interests.

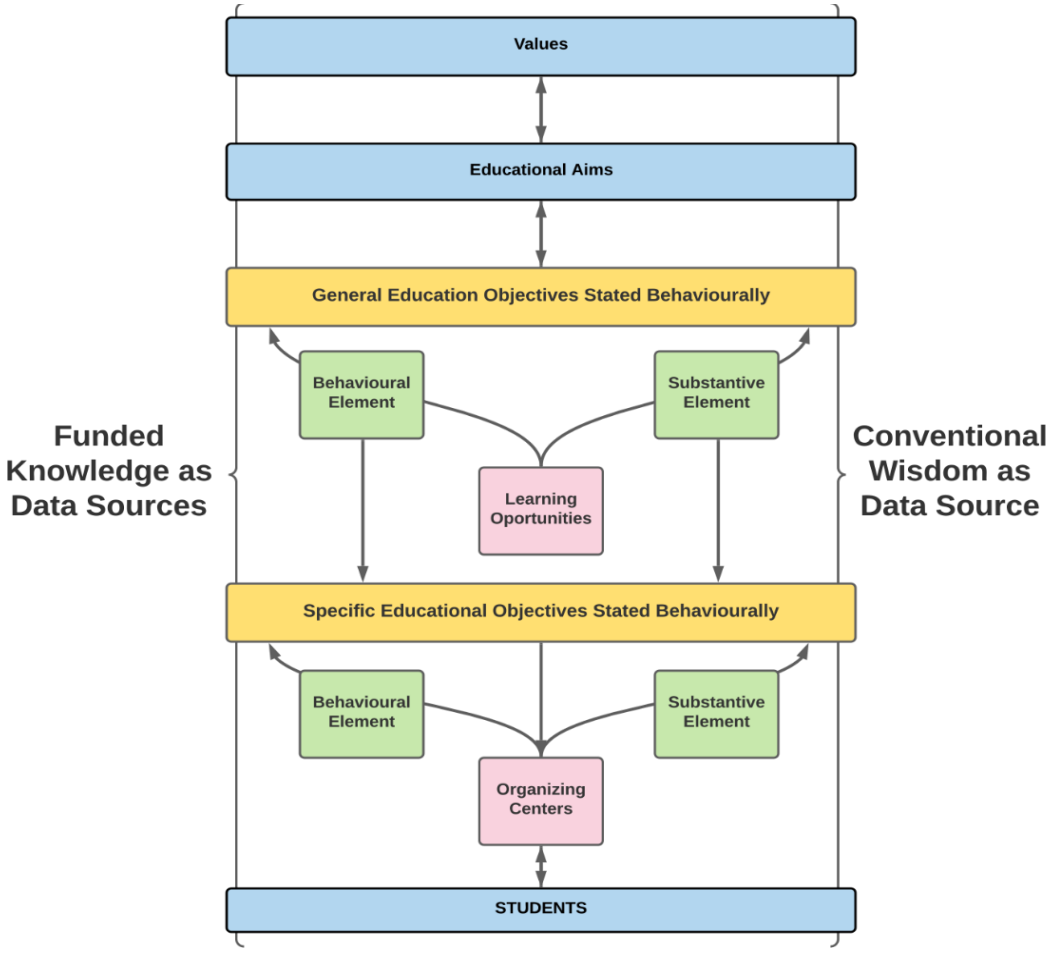


Figure 2 – John Goodlad's Model

Values

John Goodlad inclusion of values in the curriculum-development chart reflects his belief that educational systems must be driven by goals or values. He believed that education has a moral dimension, and those who teach are "moral agents." To be a professional teacher means that one is a moral agent with a moral obligation, including initiating the young into a culture.

Funded Knowledge

Funded knowledge is knowledge which is gained from research. Generally, research is heavily funded by various organizations. Information from research is used to inform educational practice in all aspects, particularly in curriculum design.

Conventional Wisdom

Conventional wisdom includes specialized knowledge within the society, for example from experts in various walks of life and 'older' people with life experiences. Students' needs and interests are also considered in the design process. Data from the various sources are then used to develop general aims of education from which general educational objectives are derived. These objectives are stated in behavioral terms. A behavioral objective has two components: a behavioral element and a substantive element. The behavioral element refers to the 'action' that a learner is able to perform, while the substantive element represents the 'content' or "substance" of the behavior.

From the general objectives, the curriculum designer identifies learning opportunities that facilitate the achievement of the general objectives. This could, for example, be specific courses of study.

The next step involves deriving specific educational objectives stated behaviorally. These are akin to instructional objectives. They are used to identify "organizing centers" which are specific learning opportunities, for example, a specific topic, a field trip, an experiment, etc.

Regarding evaluation, Goodlad proposed continuous evaluation at all stages of the design process. In the model, evaluation is represented by the double-edged arrows that appear throughout the model.

How then does Tyler's model differ with that of John Goodlad's?

Goodlad's model departs from the traditional model based on Tyler's work in several ways:

- recognition of references to scientific knowledge from research,
- use of explicit value statements as primary data sources,
- introduction of organizing centers i.e., the specific learning opportunities,
- continuous evaluation is used as a constant data source, not only as a final monitor of students' progress (formative evaluation) but also for checking each step in the curriculum planning process. Hence, the model insists upon both formative and process evaluation.

The Wheeler Model

Wheeler model of curriculum development is a type of model that indicates curriculum development is a continuous cycle. According to this model, curriculum development should be responsive to changes in the education sector and make appropriate modifications to account for these changes. Moreover, this model targets situational evaluation; this is because the context within which the curriculum decisions are taken is considered important. This practice can help make the most effective decisions.

Wheeler developed a cyclic model in reaction to criticism leveled at Ralph Tyler's model. The latter was seen as being too simplistic and vertical. By being vertical, it did not recognize the relationship between various curriculum elements.

This model includes five interconnected phases:

1. Aims, goals, and objectives
2. Learning experiences
3. Selection of content
4. Organisation and integration of learning experiences and content
5. Evaluation

Once this cycle is complete, it starts again from the first step and goes onward, continuingly improving the curriculum. Wheeler model has both advantages and disadvantages.

His cyclic proposal was therefore aimed at highlighting the interrelatedness of the various curriculum elements. It also emphasizes the need to use feedback from evaluation in redefining the goals and objectives of the curriculum.

Wheeler model takes new information into account, has a feedback mechanism, view elements of curriculum as interconnected phases, and emphasize on situational analysis. These are the main advantages of Wheeler model. However, Wheeler model is also time-consuming and can be difficult to put into practice.

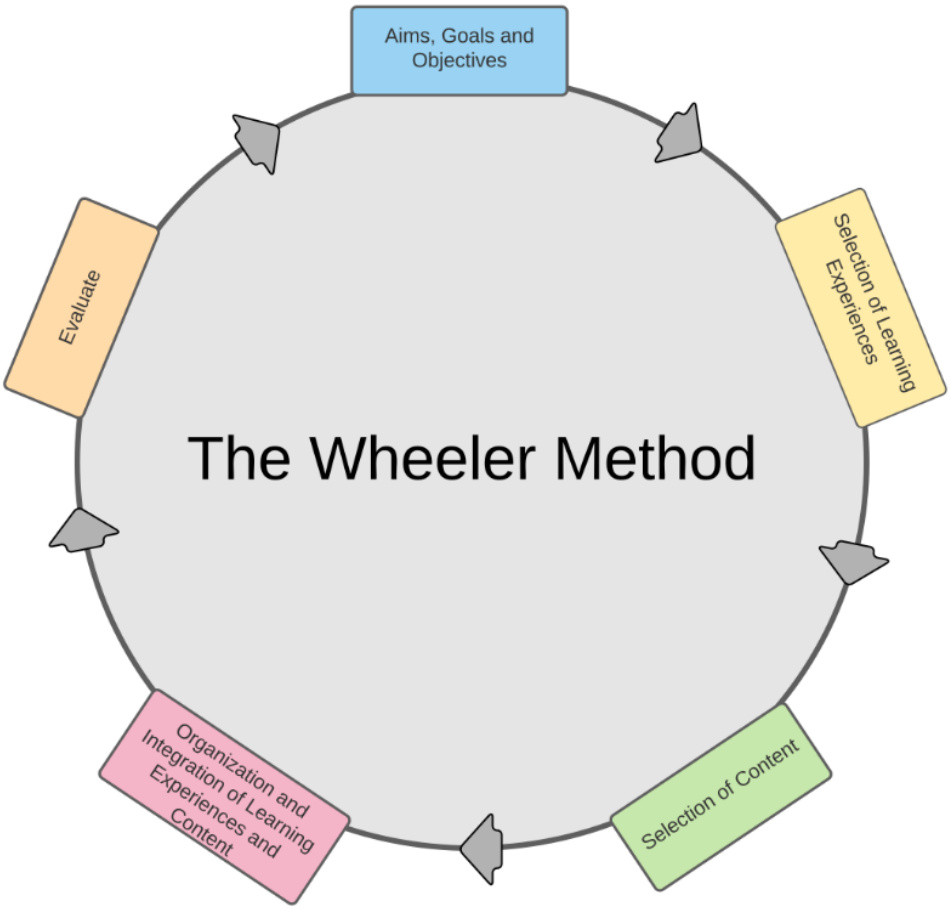


Figure 3 – The Wheeler Method

The Models of John Kerr and Hilda Taba

Other scholars who were also convinced of the 'objectives' approach to curriculum design were John Kerr and Hilda Taba. Their work is summarized in the simplified models presented in the graphic presentations that follow. Both of them emphasize the interrelatedness of the various curriculum elements.

John Kerr's Model

John Kerr, a British Curriculum specialist in the 1960s, was particularly concerned with the following issues: objectives, knowledge, school learning experiences, and evaluation. It is, however, known that new concepts, knowledge and methods of teaching invariably influence the curriculum and hence its development.

Kerr's model of the curriculum is a vast improvement on the one proposed by Tyler. It starts with the specification of objectives followed by evaluation, knowledge and learning experiences in that order. Unlike Tyler's model, the relationships in Kerr's model are not linear but interactive. The idea of cross-checking to ensure conformity with original intentions and procedures is portrayed in the model.

Kerr's model contains four elements: objective, knowledge, school learning experience, and evaluation in 1968. He saw this objective as meaning the students' expected behavioral changing after learning; these changes included perception, affection, and skills. This is reflected in the sketch below.

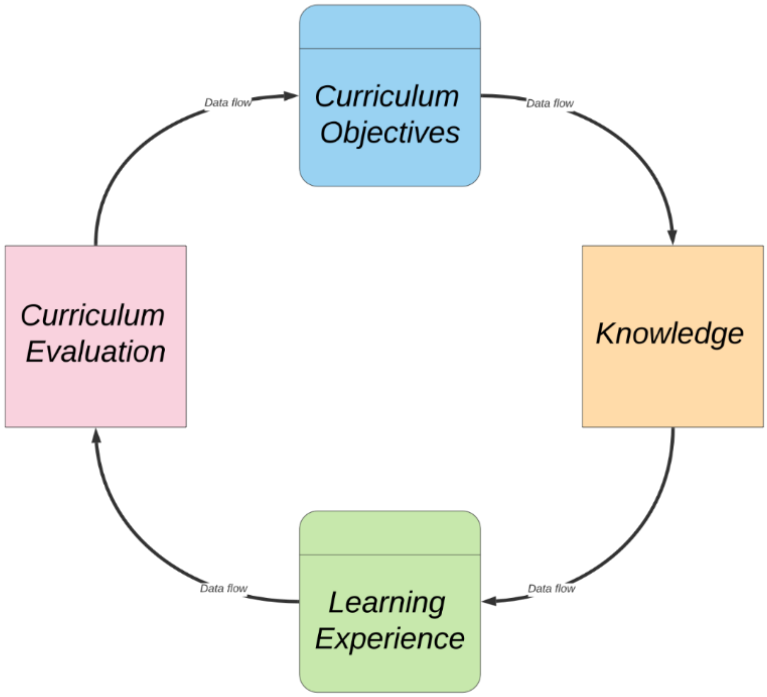


Figure 4 – John Kerr's Model

Kerr's model is in many ways similar to that of Ralph Tyler and Wheeler. The difference is the emphasis on the interrelatedness of the various components in terms of the flow of the data between each component.

Hilda Taba's Model

Hilda Taba was born in Europe and emigrated to the United States during a tumultuous time in history that had a great effect on her view of education. Hilda Taba was an architect, a curriculum theorist, a curriculum reformer, and a teacher educator. She was initially influenced by progressivists: John Dewey and Ralph Tyler, and one of her goals was to nurture the development of students and encourage them to actively participate in a democratic society. Taba's model was **inductive** rather than **deductive** in nature, and it is characterized by being a continuous process.

Taba's model emphasized concept development in elementary social studies curriculum and was used by teachers in her workshops. She was able to make connections between culture, politics, and social change as well as cognition, experience, and evaluation in curriculum development, particularly in the areas of teacher preparation and civic education. Taba's work with teachers in communities around the United States and in Europe has provided a blueprint for curriculum development that continues to be used by curriculum developers today.

Hilda Taba, on her part, was also influenced by Ralph Tyler. Her conceptual model follows. The interrelatedness of the curriculum elements from both models suggests the process is continuous.

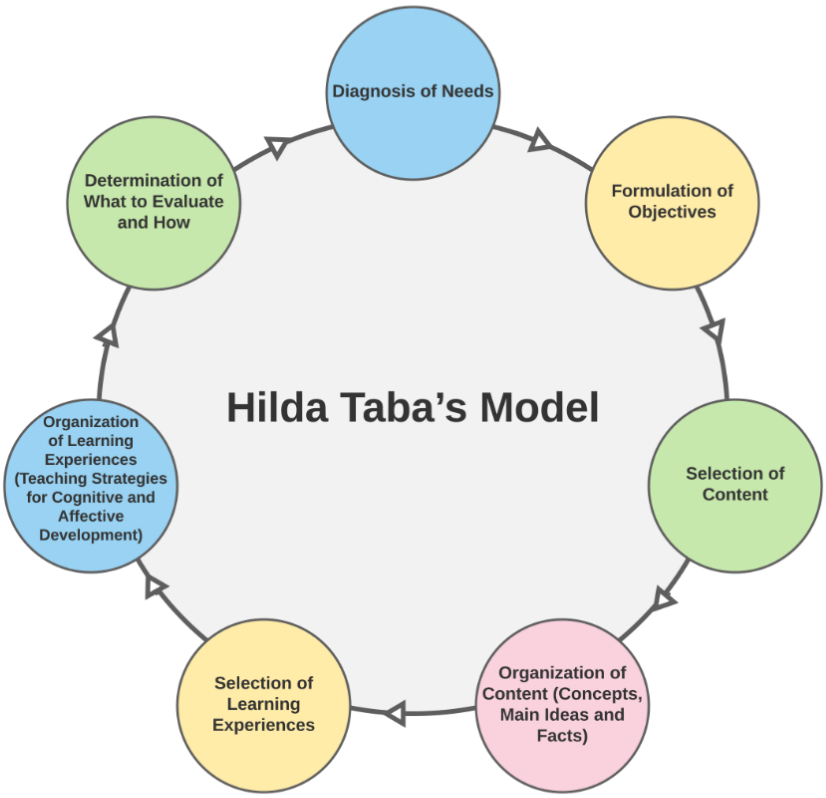


Figure 6 Hilda Taba Model

1. Diagnosis of learners need: - The teacher who is also the curriculum designer starts the process by identifying the needs of the students for whom curriculum is to be planned. For example: Majority of students are unable to think critically.

2. Formulation of Objectives: - After the teacher has identified the needs of learner that require attention, he or she Specify the objectives by which needs will be fulfilled.

3. Selection of Content: - The objective selected or created suggest the subject matter or content of the curriculum. Not only objectives and content should match, but also the validity and significance of the chosen content needs to be determined. i.e. the relevancy and significance of the content.

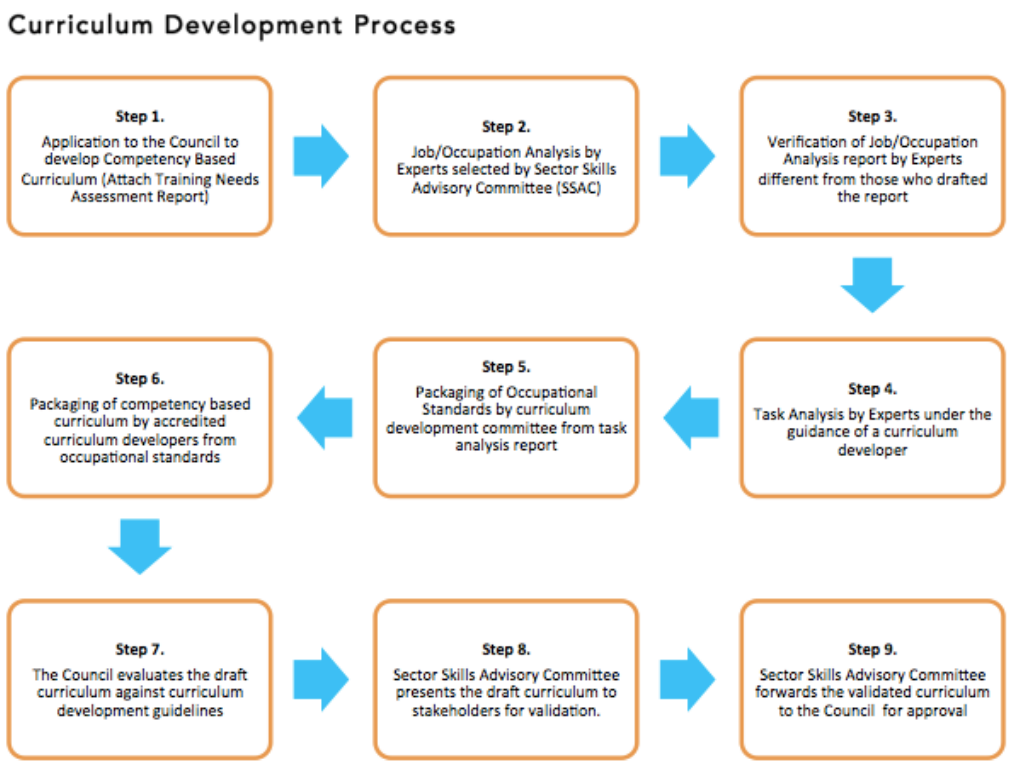
4. Organisation of content: - A teacher cannot just select content but must organise it in a Particular Sequence taking into consideration the maturity of learners, their academic achievement and their interests.

5. Selection of learning experiences: - Content must be presented to students and they must be engaged with the content. At this point teacher should select appropriate instructional methodology that will involve the students with the content.

6. Organization of learning activities: - The learning activities be organized in a sequence depending both on content sequence and learner characteristics. The teacher needs to keep in mind the student he or she will be teaching.

7. Evaluation: - The curriculum planner must determine that what objectives have been accomplished. To assess the achievement of learning objectives, evaluation procedures need to be designed.

TVET CDACC competency based curriculum design and development model



6.2.4 Factors influencing curriculum design

Several factors need to be considered when designing a curriculum. These include:

- teacher's individual characteristics,
- application of technology,
- student's cultural background and socio-economic status,
- interactions between teachers and students, and
- classroom management; among many other factors.

Question: How do the above factors affect curriculum design?

6.3 Assessment questions

1. Explain why curriculum design is important
2. In your view, what is the relationship between curriculum design and curriculum design models
3. Differentiate between subject and learner/student centered curriculum design models
4. Explain the relationship between any two of the curriculum design models discussed in this lesson.

6.4 Reference

https://oer.pressbooks.pub/curriculumessentials/chapter/curriculum-design-development-and-models%E2%80%AF%E2%80%AF-planning-for-student-learning-there-is-always-a-need-for-newly-formulated-curriculum-models-that-address-contemporary-circumstance-an/